

Controls, Fixed Effects, and Causality

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Quantitative Reasoning II · INF 1220 · University of Austin

Lesson 10

Controls

A variable added to a model so that it is held constant is called a **control**.

Papers say the same thing several ways:

- “controlling for age and quality”

Which predictor is the control depends on the question.

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- “controlling for age and quality”
- “adjusted for age and quality”
- “age and quality were included as covariates”

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A New Setting

- 500 largest US companies by stock market value in 2019

Gross margin: profit as a percent of sales after the cost of making or buying the product.

R&D share: spending on research and development as a percent of sales.

Dividing by sales lets companies of very different sizes be compared: \$1 billion of R&D is a lot for a company with \$3 billion in sales and little for Apple, with \$260 billion.

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- each row is a company's financial data for 2019

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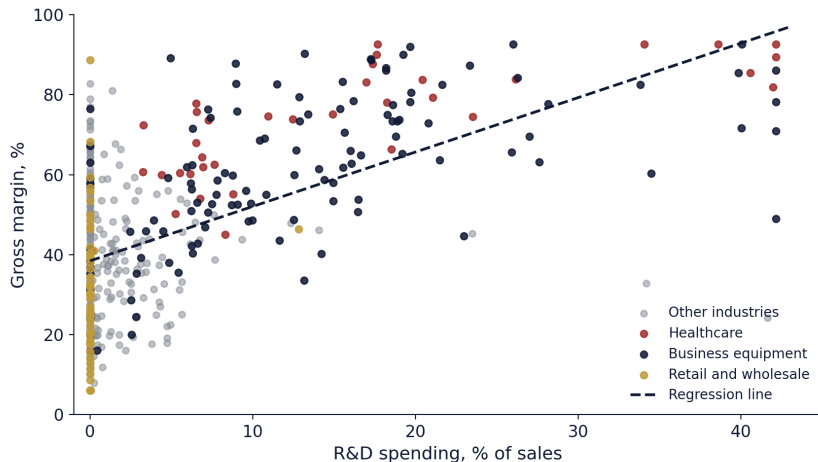
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Some of the Companies

	Industry	Gross margin	R&D share
Pfizer	Healthcare	83.1%	17.0%
Microsoft	Business equipment	75.1%	13.4%
Intel	Business equipment	73.3%	18.6%
Coca-Cola	Consumer nondurables	64.4%	0.0%
Procter & Gamble	Chemicals	53.6%	2.7%
Apple	Business equipment	42.2%	6.2%
Caterpillar	Manufacturing	35.4%	3.1%
Walmart	Retail and wholesale	26.4%	0.0%
Ford	Consumer durables	18.0%	4.7%
Exxon Mobil	Energy	17.4%	0.5%

Do High R&D Companies Have Bigger Margins?



Gross Margin on R&D Share

Dependent variable: gross margin (%)				
	Coef.	Std. Err.	<i>t</i>	<i>p</i> -value
Constant	38.45	0.863	44.54	<0.001
R&D share	1.36	0.083	16.30	<0.001
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The R&D coefficient is statistically significant at the 0.001 level – the coefficient is reliably nonzero.

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A good t -statistic has $|t| > 2$. The bigger $|t|$, the stronger the evidence that the coefficient is not zero.

Averages by Industry

	Companies	Gross margin	R&D share
Healthcare	43	65.4%	13.4%
Business equipment	118	61.8%	13.3%
Manufacturing	55	33.7%	2.6%
Retail and wholesale	60	32.1%	0.2%
Utilities	36	35.2%	0.0%

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Does our coefficient really describe companies, or is it heavily influenced by industries?

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All without measuring!

Model with Industry Fixed Effects

$$\begin{aligned}\widehat{\text{gross margin}} &= 49.79 + 0.90 \times \text{R\&D share} \\ &\quad + 3.55 \times \text{healthcare} - 17.86 \times \text{retail} \\ &\quad - 18.40 \times \text{manufacturing} + \dots\end{aligned}$$

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healthcare = 1 if the company is in healthcare and 0 if not. Retail and manufacturing work the same way.

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The model includes an indicator like these for every industry except business equipment, the base category: ten indicators for eleven industries.

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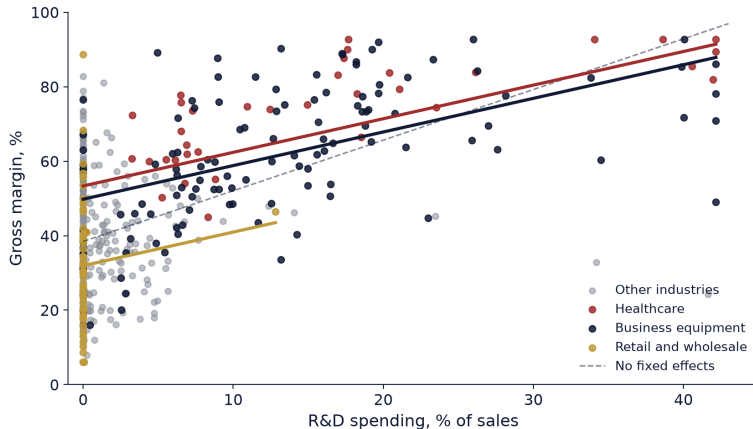
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The R&D slope is 0.90 in every industry. Turning an indicator on adds a fixed amount to the intercept: +3.55 for healthcare, making it 53.34, and -17.86 for retail, making it 31.93.

Model with Industry Fixed Effects



Dashed: no fixed effects.

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Outcome: gross margin, %	(1)
R&D share, %	1.36
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R^2	0.35
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Model with Industry Fixed Effects

Outcome: gross margin, %	(1)	(2)
R&D share, %	1.36	0.90
Industry fixed effects	No	Yes
R^2	0.35	0.47
n	500	500

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(1) to (2): holding industry constant, a one-percentage-point increase in R&D share is associated with a 0.90-percentage-point increase in gross margin. The slope falls by a third. Column (2) has 12 coefficients; the table prints one of them and a “Yes,” which is how papers report fixed effects.

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A control changes a slope only when it is related both to that predictor and to the outcome.

Model with Industry Fixed Effects: Software Output

=====				
Dep. variable: gross_margin		R-squared	0.468	
Model: OLS		Adj. R-squared	0.456	
Observations: 500		F(11, 488)	39.04	

	coef	std err	t	P> t

const	49.7905	1.9238	25.88	0.000
R&D share	0.9029	0.0987	9.15	0.000
Chemicals	-8.1961	3.7249	-2.20	0.028
Consumer durables	-22.5040	4.1691	-5.40	0.000
Consumer nondurables	-5.2570	3.1375	-1.68	0.094
Energy	-12.5225	3.6084	-3.47	0.001
Healthcare	3.5547	2.7296	1.30	0.193
Manufacturing	-18.4042	2.7141	-6.78	0.000
Other	-14.2642	2.5865	-5.51	0.000
Retail and wholesale	-17.8603	2.7492	-6.50	0.000
Telecom	0.7134	4.1779	0.17	0.864
Utilities	-14.5652	3.1974	-4.56	0.000
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- The industries as defined are broad. Business equipment puts Apple in with chip makers and software companies
 - one might consider different specifications of industry
- A broad group holds less constant.

Reporting Fixed Effects

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Family	everything siblings share

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Every regression so far describes an association (even those with fixed effects), not a causal relationship.

The causal version of our question today

Today, we asked: Do higher R&D companies have bigger margins?

The causal version of this question is: If a company spent more on R&D, would its gross margin rise?

The comparison we want is the same company with and without the extra R&D, but that's not possible.

The next best is two groups of companies that are alike in everything that affects margins, except R&D.

Confounders and Lurking Variables

Industry is a **confounder**. It is related to R&D share and to gross margin, so its effect is mixed into the comparison.

A **lurking variable** is a confounder left out of the model. Industry was a lurking variable in column (1).

The bias a lurking variable leaves in a coefficient is called **omitted variable bias**.

Why including fixed effects does not make the comparison causal

Industry fixed effects hold a great deal constant, but there are still many differences between companies that might be correlated with both R&D and margins.

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- We can only control for things that we can measure in some way
- Reverse causality here is possible: high margins may allow a business to spend more on R&D

How can we make causal claims?

To claim a causal relationship we need to compare groups that differ *only* in the treatment

The **treatment** is the variable whose effect we want to know (e.g. increase in R&D spending, a new drug, a tax policy)

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- **Before and after:** follow the same individuals over time, so each is its own comparison

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